Geometry & Measurement

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Angle facts

Two angles on a straight line measure $3x$ and $2x + 30$. Find x.

2 [Foundational] • Target: Angle facts

Find the angle vertically opposite a 65° angle.

3 [Foundational] • Target: Angle facts

Three angles around a point measure 110°, 95°, and x. Find x.

4 [Foundational] • Target: Angle facts

Two angles are complementary (sum to 90°). One measures 37°. Find the other.

5 [Foundational] • Target: Parallel lines and transversals

Two parallel lines are cut by a transversal, creating an angle of 70°. Find its corresponding angle.

6 [Foundational] • Target: Parallel lines and transversals

In the same setup, find the co-interior (allied) angle to a 70° angle.

7 [Foundational] • Target: Parallel lines and transversals

Find the alternate interior angle to a 55° angle.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

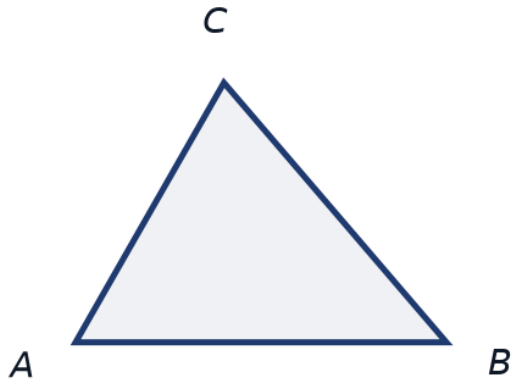
8 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x.

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9 [Intermediate] • Target: Triangle angle sum

Find the third angle of the triangle shown, given the two marked angles.



10 [Intermediate] • Target: Triangle angle sum

An isosceles triangle has two equal angles of 72° each. Find the third angle.

11 [Intermediate] • Target: Triangle angle sum

In a right triangle, one of the non-right angles is 35°. Find the other non-right angle.

12 [Intermediate] • Target: Triangle angle sum

A triangle has three equal angles. Find each angle, and name this type of triangle.

13 [Intermediate] • Target: Angle sums of polygons

Find the sum of the interior angles of a hexagon (6 sides).

14 [Intermediate] • Target: Angle sums of polygons

Find the sum of the interior angles of a pentagon (5 sides).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

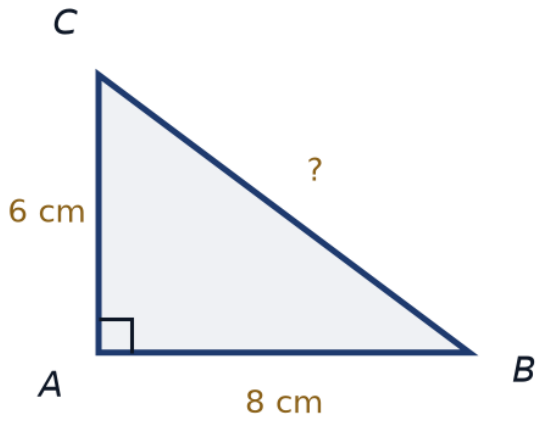
15 [Challenging] • Target: Angle sums of polygons • Cross-Lesson: 5.01 Quantitative Comparisons

A regular polygon has an interior angle sum of $1{,}080^\circ$. How many sides does it have?

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16 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 6.01 Sequences & Patterns

Find the hypotenuse of the right triangle shown.



17 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 7.01 Data Analysis & Probability

A right triangle has hypotenuse 13 and one leg 5. Find the other leg.

18 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 8.01 Word Problems & Reasoning

A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?

19 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 1.01 Arithmetic & Number Sense

A right triangle has legs 9 and 12. Find the hypotenuse.

20 [Challenging] • Target: Perimeter and area of rectangles and squares • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

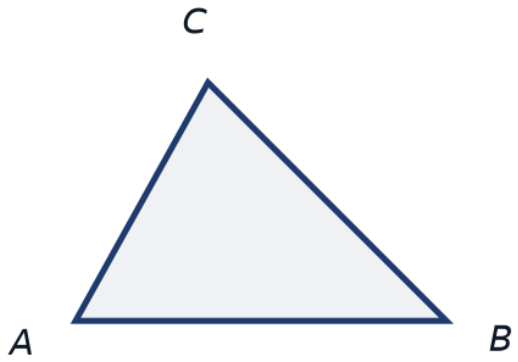
A rectangle has length 12 cm and width 7 cm. Find its perimeter and area.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Triangle angle sum

Find the third angle of the triangle shown.



a. 116°

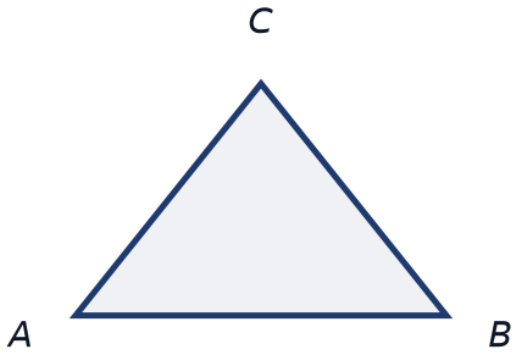
b. 64°

c. 54°

d. 44°

22 [Foundational] • Target: Triangle angle sum

The isosceles triangle shown has two equal base angles of 70° each. Find the apex angle.



a. 55°

b. 70°

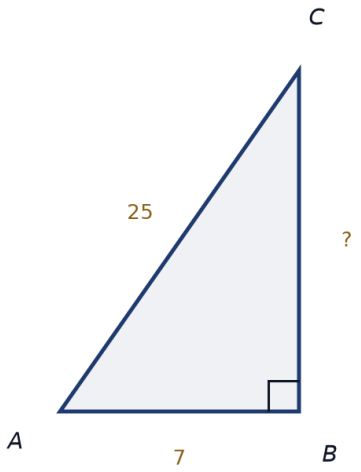
c. 40°

d. 110°

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23 [Foundational] • Target: The Pythagorean theorem

Find the missing leg of the right triangle shown.



a. 26

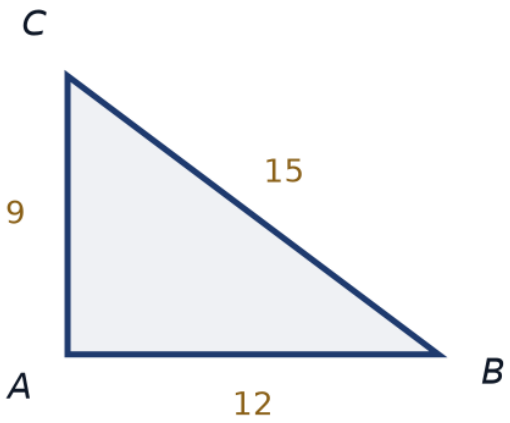
b. 18

c. 24

d. 20

24 [Foundational] • Target: The Pythagorean theorem

A triangle has sides 9 cm, 12 cm, and 15 cm, as shown. Is it a right triangle?



a. Only if it is also isosceles

b. Yes — a right triangle

c. Cannot be determined

d. No — not a right triangle

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25 [Foundational] • Target: Perimeter and area of rectangles and squares

Find the perimeter and area of the rectangle shown.

a. $P = 48$ cm, $A = 135$ cm^2

b. $P = 48$ cm, $A = 24$ cm^2

c. $P = 135$ cm, $A = 48$ cm^2

d. $P = 24$ cm, $A = 135$ cm^2

26 [Foundational] • Target: Perimeter and area of rectangles and squares

Find the perimeter and area of the square shown.

a. $P = 44$ cm, $A = 110$ cm^2

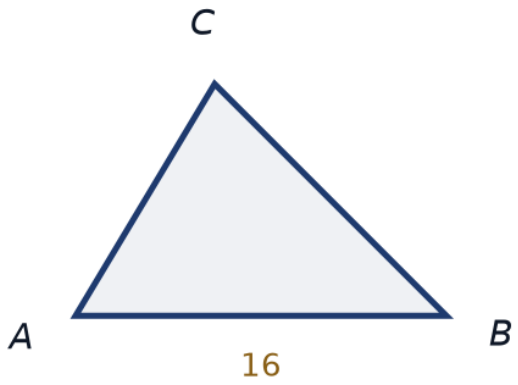
b. $P = 44$ cm, $A = 121$ cm^2

c. $P = 22$ cm, $A = 121$ cm^2

d. $P = 121$ cm, $A = 44$ cm^2

27 [Foundational] • Target: Area of triangles and parallelograms

The triangle shown has base 16 cm and height 10 cm. Find its area.



a. 160

b. 26

c. 80

d. 40

28 [Foundational] • Target: Area of triangles and parallelograms

The parallelogram shown has base 18 cm and height 7 cm. Find its area.

a. 252

b. 126

c. 63

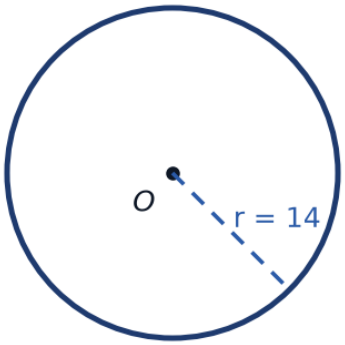
d. 25

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Circles: circumference and area**

Find the circumference of the circle shown, using $\pi = \frac{22}{7}$.



a. 44

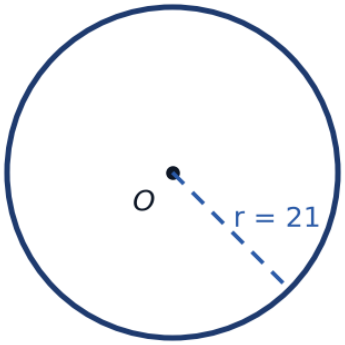
b. 616

c. 88

d. 176

30 **[Intermediate]** • **Target: Circles: circumference and area**

Find the area of the circle shown, using $\pi = \frac{22}{7}$.



a. 132

b. 2,772

c. 693

d. 1,386

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31 [Intermediate] • Target: Angle facts

Two angles are supplementary. One is three times the other. Find the smaller angle.

a. 60°

b. 30°

c. 135°

d. 45°

32 [Intermediate] • Target: Angle facts

An angle is four times its complement. Find the angle.

a. 54°

b. 18°

c. 72°

d. 90°

33 [Intermediate] • Target: Angle facts

Two angles on a straight line measure $(4x + 10)^\circ$ and $(2x + 20)^\circ$. Find x.

a. 20

b. 35

c. 15

d. 25

34 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines are cut by a transversal. One angle is $(3x + 15)^\circ$ and its corresponding angle is $(5x - 25)^\circ$. Find x.

a. 10

b. 8

c. 30

d. 20

35 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines cut by a transversal create co-interior angles of $(2x + 10)^\circ$ and $(3x - 30)^\circ$. Find x.

a. 30

b. 40

c. 50

d. 34

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36 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines are cut by a transversal. One angle is 63°. Find its alternate exterior angle.

a. 153° **b.** 27°

c. 63° **d.** 117°

37 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines are cut by a transversal. One angle and its co-interior angle differ by 40° and sum to 180°. Find the larger angle.

a. 110° **b.** 70°

c. 140° **d.** 90°

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Triangle angle sum • Cross-Lesson: 8.01 Word Problems & Reasoning

A triangle has three angles in the ratio $2:3:4$. Find the largest angle.

a. 90° **b.** 100°

c. 80° **d.** 60°

39 [Challenging] • Target: Triangle angle sum • Cross-Lesson: 1.01 Arithmetic & Number Sense

In a right triangle, one non-right angle exceeds the other by 26°. Find the larger non-right angle.

a. 32° **b.** 58°

c. 64° **d.** 48°

40 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A rectangular field is 24 m long and 18 m wide. Find the length of its diagonal.

a. 30 **b.** 36

c. 42 **d.** 26

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41 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 3.01 Algebra Essentials

A 15 m ladder leans against a wall with its foot 9 m from the wall's base. How high up the wall does the ladder reach?

a. 13

b. 10

c. 12

d. 18

42 [Challenging] • Target: Area of triangles and parallelograms • Cross-Lesson: 5.01 Quantitative Comparisons

A triangle has area 84 cm^2 and height 12 cm. Find its base.

a. 14

b. 1,008

c. 28

d. 7

43 [Challenging] • Target: Area of triangles and parallelograms • Cross-Lesson: 6.01 Sequences & Patterns

A parallelogram has area 150 cm^2 and base 15 cm. Find its height.

a. 20

b. 10

c. 2,250

d. 5

44 [Challenging] • Target: Circles: circumference and area • Cross-Lesson: 7.01 Data Analysis & Probability

A circle has area 144π cm^2. Find its radius.

a. 6

b. 24

c. 12

d. 72

45 [Challenging] • Target: Composite figures • Cross-Lesson: 8.01 Word Problems & Reasoning

A garden is L-shaped: an 18 m by 8 m rectangle with a 6 m by 3 m rectangle removed from one corner. Find its area.

a. 108

b. 126

c. 162

d. 138

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